

Imprecise probability and valid statistical inference

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- Statistics aims to give reliable/valid uncertainty quantification about unknowns based on data, models, etc.
- Two dominant schools of thought:
 - *frequentist*
 - *Bayesian*
- Dichotomy is bad: creates confusion, masks opportunities
- Goals are the same, but there's an obvious spectrum:
 - the more I'm willing to assume, the more precise I can be
 - the less I'm willing to assume, the less precise I can be
- Different points on the spectrum (even the extremes) shouldn't require entirely different theories...

- Start with a sufficiently general UQ framework.
- An advantage of Bayes: (*precise*) probabilities for everything.
- *Imprecise probability* is more general.
- Actually, it's too general without statistical constraints.
- My goals are:
 - define what valid/reliable means in this framework
 - see what mathematical form is needed to achieve validity
 - practical, theoretical, and computational implications?
- Partially done — I'll tell you what I know and what's next.

- Problem setup
- Uncertainty quantification via inferential models (IMs)
- Validity property
- Main results:
 - precise probability isn't valid
 - possibility/consonant plausibility can be valid
 - possibilistic characterization of frequentism
- Constructing a valid IM
- Remarks, open questions, etc.



- Don Fraser passed away on Dec 21st, 2020, age 95.
- Recognized and wrote about the limitations of ordinary probability in the context of statistical inference.³
- His work made a tremendous impact on the field of statistics and on me personally.

³e.g., "Is Bayes just quick and dirty confidence?" *Statist. Sci.*, 2011.

Statistical inference problem

- Observable data: Y in a sample space $\mathbb{Y}$.
- Statistical model: a collection of probability distributions

$$\mathcal{P} = \{P_{\vartheta} : \vartheta \in \Theta\}.$$

- θ is the unknown true value; ϑ is a generic value.
- One end of the spectrum: *no prior information*.
- Goal is to learn from data, y , about θ :
 - uncertainty quantification
 - think “data-dependent belief assignments”

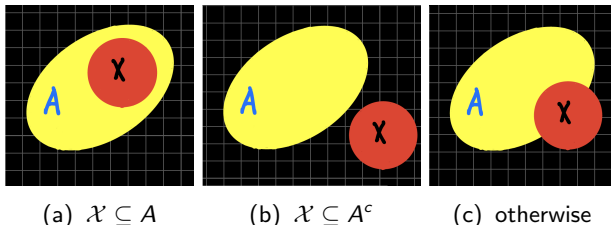
Definition.

An *inferential model* (IM) is a function whose input is $(y, \mathcal{P}, \dots)$ and whose output is a capacity^a $\underline{\Pi}_y : 2^\Theta \rightarrow [0, 1]$ such that,

$$\underline{\Pi}_y(A) = \text{degree of belief in } "\theta \in A," \text{ given } y, \quad A \subseteq \Theta$$

^amonotone, continuous set function

- The “...” allows for other inputs, e.g., prior information.
- Examples include: Bayes, fiducial, Dempster–Shafer,
- Note: the capacity could be *non-additive*.



- Example illustrates non-additivity induced by a *random set* $\mathcal{X}$
 - if $\underline{\Pi}(A) = P(\mathcal{X} \subseteq A)$,
 - then $\underline{\Pi}(A) + \underline{\Pi}(A^c) \leq 1 \leftarrow \text{non-additive!}$
- Under non-additivity, the *dual* to $\underline{\Pi}_Y$ is $\overline{\Pi}_Y(A) = 1 - \underline{\Pi}_Y(A^c)$
 - Fact: $\underline{\Pi}_Y(A) \leq \overline{\Pi}_Y(A)$
 - $\overline{\Pi}_Y(A)$ represents the plausibility of A .

- IM output is a pair $(\underline{\Pi}_y, \overline{\Pi}_y)$.
- Just like with probabilities, these are subjective or personal.
- Behavioral interpretation is via gambling:
 - $\underline{\Pi}_y(A)$ is my max buying price for $\$1\{\theta \in A\}$
 - $\overline{\Pi}_y(A)$ is my min selling price for $\$1\{\theta \in A\}$
- Coherence properties:
 - constraints on $A \mapsto \underline{\Pi}_y(A)$ protect me from sure loss
 - my internal sanity check
- More is needed to make my beliefs real-world relevant.
- “More” = statistical constraints.

- What kind of statistical constraints?
- *Basic principle:* if $\underline{\Pi}_Y(A)$ is large, infer A .
- Reid & Cox:
it is unacceptable if a procedure. . . of representing uncertain knowledge would, if used repeatedly, give systematically misleading conclusions.
- We don't want $\underline{\Pi}_Y(A)$ to be large if A is false.
- Idea: require that $y \mapsto \underline{\Pi}_y(\cdot)$ satisfy
$$\{\theta \notin A \text{ and } Y \sim P_\theta\} \implies \underline{\Pi}_Y(A) \text{ tends to be small.}$$

Definition.

An IM $(y, \mathcal{P}, \dots) \mapsto \underline{\Pi}_y$ is *valid* if

$$\sup_{\theta \notin A} P_{\theta} \{ \underline{\Pi}_Y(A) > 1 - \alpha \} \leq \alpha, \quad \text{for all } A \subseteq \Theta, \alpha \in [0, 1]$$

- Validity controls the frequency at which the IM assigns relatively high beliefs to false assertions.
- There's an equivalent statement in terms of $\overline{\Pi}_y$:

$$\sup_{\theta \in A} P_{\theta} \{ \overline{\Pi}_Y(A) \leq \alpha \} \leq \alpha, \quad \text{for all } A \subseteq \Theta, \alpha \in [0, 1].$$

- “For all A ” is important, e.g., for marginalization.

Theorem.

If $(\underline{\Pi}_Y, \overline{\Pi}_Y)$ are valid, then derived procedures control error rates:

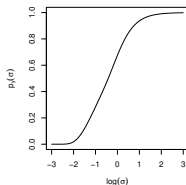
- “reject $H_0 : \theta \in A$ if $\overline{\Pi}_Y(A) \leq \alpha$ ” is a size α test,
- the $100(1 - \alpha)\%$ plausibility region $\{\vartheta : \overline{\Pi}_Y(\{\vartheta\}) > \alpha\}$ has coverage probability $\geq 1 - \alpha$.

- IM validity $\implies$ usual frequentist validity
- Connection is mutually beneficial:
 - IMs help with interpretation of frequentist output
 - calibration makes IM's $(\underline{\Pi}_Y, \overline{\Pi}_Y)$ real-world relevant

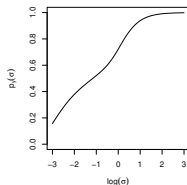
- What mathematical form does a valid IM take?
 - 1 *if IM is additive, then it's not valid*
 - 2 *if $\bar{\Pi}_Y$ is a consonant plausibility function, aka a possibility measure, then it can be valid.*
- There's a sort of converse to **Main Result 2**:
 - 3 *procedures with frequentist error rate control correspond to valid IMs that are possibility measures.*
- **2 + 3** $\implies$ frequentist $\equiv$ valid IMs as possibility measures.

■ Satellite conjunction analysis:

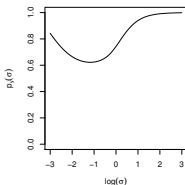
- Orbiting satellite could collide with another object.
- To avoid this, analysts compute a *non-collision probability*.⁴
- Satellite judged to be safe if non-collision probability is high.
- *Noisier data makes non-collision probability large*⁵



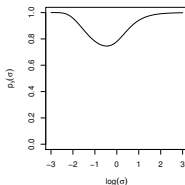
(d) Close



(e) Kinda close



(f) Kinda far



(g) Far

⁴Details in Balch, M., and Ferson (2019), [arXiv:1706.08565](https://arxiv.org/abs/1706.08565).

⁵M. (2019), researchers.one/articles/19.02.00001

- *False confidence*: Hypotheses tending to be assigned high probability even if data don't support them.
- Probabilities suffer from false confidence, *not valid*.

False confidence theorem (Balch, M., and Ferson 2019).

Let Π_Y be a probability on Θ , depending on data Y . Then, for any $\alpha \in (0, 1)$ and any $\theta \in \Theta$, there exists $A \subset \Theta$ such that

$$A \not\ni \theta \quad \text{and} \quad P_\theta\{\Pi_Y(A) > 1 - \alpha\} > \alpha.$$

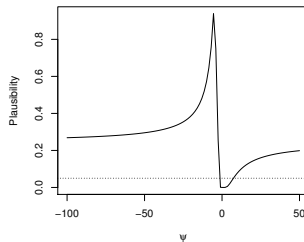
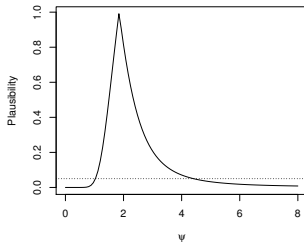
- This result is not too surprising:
 - probabilities are too precise
 - data may not be informative enough to reliably support that level of precision
 - e.g., mean vector vs. a weird function of it
- This is not the consequence of using a “bad prior,” etc., it’s entirely due to additivity.
- Non-additivity creates imprecision:
 - opportunity to be less committal
 - lower our risk of being wrong (cf. Reid & Cox)
 - help us avoid false confidence.
- What’s the “right” kind of imprecision?

Main result 2

- Let $\pi_y : \Theta \rightarrow [0, 1]$ be a function with $\sup_{\vartheta} \pi_y(\vartheta) = 1$ and define

$$\overline{\Pi}_y(A) = \sup_{\vartheta \in A} \pi_y(\vartheta), \quad A \subseteq \Theta.$$

- Then $\overline{\Pi}_y$ is a *consonant plausibility function*, or *possibility measure*
- and π_y is its *plausibility contour*.
- $\underline{\Pi}_y(A) := 1 - \overline{\Pi}_y(A^c) < \overline{\Pi}_y(A) \leftarrow$ non-additive!



- Frequentist procedures are based on p-values:
 - $(y, \vartheta) \mapsto \pi_Y(\vartheta)$, small values imply y and ϑ disagree
 - If $Y \sim P_\theta$, then $\pi_Y(\theta) \sim \text{Unif}(0, 1)$.
- Comparison:
 - p-value + additivity $\rightarrow$ not valid.
 - p-value + consonance $\rightarrow$ valid.⁶

Theorem.

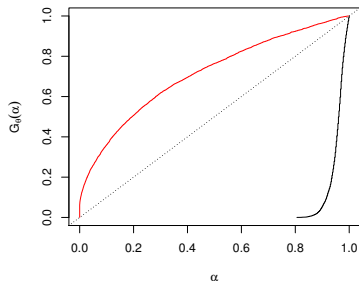
If the p-value π_Y meets the conditions of a plausibility contour, then the IM with $\bar{\Pi}_Y(A) = \sup_{\vartheta \in A} \pi_Y(\vartheta)$ is valid. Moreover,

$$P_\theta\{\bar{\Pi}_Y(A) \leq \alpha \text{ for some } A \ni \theta\} \leq \alpha.$$

⁶M. (2021), researchers.one/articles/21.01.00002

Main result 2, cont.

- $Y \sim P_\theta = N_2(\theta, I)$
- Two IMs:
 - $\Pi_y = N_2(y, I)$.
 - $\underline{\Pi}_y$, p-value + consonance
- Target is the ratio $\phi = \theta_1/\theta_2$.
- Let $A = \{\vartheta : \phi(\vartheta) \leq 9\}$.
- Suppose $\theta = (1, 0.1)$, so that $\phi = 10$ and A is *false*.
- Then $\Pi_y(A)$ shouldn't be big...



- CDFs of $\Pi_Y(A)$ & $\underline{\Pi}_Y(A)$
 - big $\rightarrow$ not valid
 - not big $\rightarrow$ valid

- Combining previous results, we get

$$\{\text{consonant IMs}\} \subseteq \{\text{frequentist}\}.$$

- But to put “frequentist” on the spectrum of valid IMs, I need the opposite inclusion too:

$$\{\text{consonant IMs}\} \supseteq \{\text{frequentist}\}.$$

- Challenge is that the “frequentist approach” has no rules, so the right-hand side contains all sorts of things.
- But I can get basically what I need...

Theorem.

Let $\{C_\alpha : \alpha \in [0, 1]\}$ be a family of confidence regions for $\phi = \phi(\theta)$ that satisfies the following properties:

Coverage. $\inf_{\theta} P_{\theta}\{C_{\alpha}(Y) \ni \phi(\theta)\} \geq 1 - \alpha$ for all α ;

Nested. if $\alpha \leq \beta$, then $C_{\beta} \subseteq C_{\alpha}$;

Compatible.

There exists a valid IM for θ , a *possibility measure*, whose derived marginal plausibility regions $C_{\alpha}^*(y)$ for ϕ satisfy

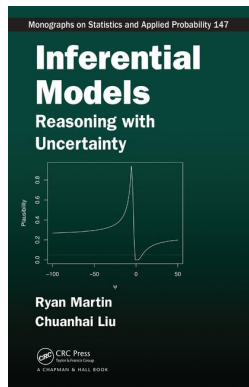
$$C_{\alpha}^*(y) \subseteq C_{\alpha}(y) \quad \text{for all } (y, \alpha) \in \mathbb{Y} \times [0, 1].$$

M. (2021), researchers.one/articles/21.01.00002

- The compatibility condition is too messy to talk about
 - detailed explanation in the paper
 - I haven't found a “not-weird” example where it fails
- Note: confidence region is for $\phi(\theta)$ but valid IM is for θ .
- Analogous result for hypothesis tests.
- Take-away message:
 - anytime you use/teach [insert favorite method]
 - or you prove a new method controls frequentist error rates,
 - there's a plausibility contour and a valid IM, *which you have access to*, doing it's thing behind the scenes

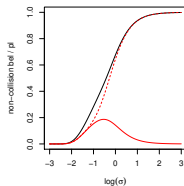
- Where does a valid IM come from?
- Three strategies for constructing a valid IM:
 - (nested) random sets on an auxiliary space
 - possibility measure on an auxiliary space
 - “Rao–Blackwellization”
- Just a super high-level intro to these...

- $Y = a(\theta, U)$, $U \sim P_{\text{known}}$.
- If we could observe U , then inference on θ is trivial.
- Idea: use a suitable random set to “guess” the unobserved U .
- Push random set through to a data-dependent possibility measure on θ -space.
- First validity theorems!

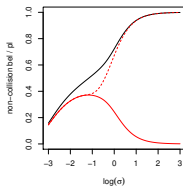


Valid IM construction, cont.

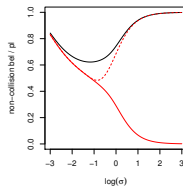
- Satellite collision illustration: $A = \{\text{non-collision}\}$.
 - $\Pi_Y(A)$ (black)
 - $\underline{\Pi}_Y(A)$ (red) and $\overline{\Pi}_Y(A)$ (red, dashed).
- Gap between $\underline{\Pi}_Y(A)$ and $\overline{\Pi}_Y(A)$ is increasing in σ .
- $\Pi_Y(A) \approx \overline{\Pi}_Y(A) \implies$ the “precision” in Π_Y is *artificial*.



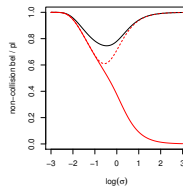
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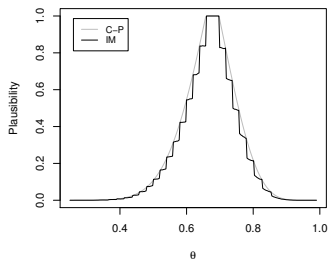
(k) Far

- Possibility measures on an auxiliary space.⁷
- Similar to the use of (nested) random sets mentioned above.
- Might be more straightforward in some cases.
- There are some existing “optimality” results.
- Hasn't been carefully explored yet...

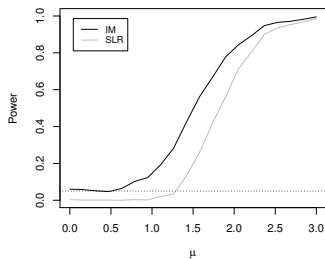
⁷Liu and M. (2020), researchers.one/articles/20.08.00004

Valid IM construction, cont.

- Proof of Main Result 3 is *constructive*
- That is, the proof gives a sort of “algorithm”
 - input: a given procedure
 - output: a valid IM that’s at least as good (Rao–Blackwell).



(l) Binomial plausibility



(m) Power, mixture model

- Interpretation of inferential output is always non-trivial.
 - Bayes feels easier because probability is familiar.
 - Frequentist felt harder because, e.g., p-values were not well-defined mathematical objects.
- Mathematical facts:
 - p-values measure the plausibility of the null hypothesis
 - confidence regions are collections of parameter values that are sufficiently plausible
- I teach this interpretation to my students:

"I wanted to let you know that your discussion of plausibility in ST503 made it easier for me to explain what a p-value is when I was interviewing. Thank you for the new perspective on the subject!"

- Computation with imprecise probs is generally non-trivial.
- They're more complex/nuanced objects than probabilities.
- However, possibility measures are the simplest.
- Computation requires:
 - evaluate plausibility contour, π_y , may require Monte Carlo
 - do optimization $\bar{\Pi}_y(A) = \sup_{\vartheta \in A} \pi_y(\vartheta)$.
- For standard problems, this is relatively easy.⁸⁹
- For more complex, high-dim problems, it's definitely not impossible — an interesting/challenging research problem?

⁸Joyce Cahoon's PhD thesis and associated papers.

⁹Syring and M. (2021), [arXiv:2103.02659](https://arxiv.org/abs/2103.02659)

- Beyond inference, one might be concerned with prediction.
- All of what I said above applies to prediction too.
- Practical challenge:
 - “model-free” prediction is the goal
 - but the above machinery is model-based
- What about a valid *nonparametric* IM?
- Turns out there's a close connection between the imprecise probability stuff here and *conformal prediction*.¹⁰
- “Conformal + consonance” leads to a powerful and valid nonparametric IM for prediction.

¹⁰Cella and M. (2021), researchers.one/articles/20.01.00010

What's next?

- Model-free inference is an interesting problem.¹¹
- Multinomial example would shed light on valid IMs for general nonparametric problems.
- The “spectrum” I mentioned above is still unclear.
- Imprecise prob is flexible, allows for *partial prior info*.
- High-dim problems are my motivation:
 - structural assumptions, e.g., sparsity, parsimony, etc.
 - this is just partial prior information
- How to use imprecise prob to incorporate this in a way that's both valid and efficient along the spectrum?

¹¹Leo Cella's PhD thesis and associated papers.

- We've always known that statistical reasoning is imprecise.
- Now we can see this in the math:
 - validity is crucial to the logic, and
 - imprecise probability is necessary for validity.
- *Plausibility/possibility* is especially important:
 - for the theory
 - for the interpretation
 - for the computation
- Hopefully can capture the spectrum from no prior info to complete prior info, and a corresponding notion of validity.
- New and exciting territory.

References



Thanks for your attention!

`www4.stat.ncsu.edu/~rgmarti3`

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...

We're excited to be working with the society on imprecise probability theory and applications (SIPTA) on hosting their 2021 virtual conference [@researchersone](#)



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Registration and management of virtual conference will be provided by Researchers One [researchers.one](#)